

Example 11 : Consider again the channels of Example 1 & Example 2, together with the given input ensemble. Find how many bits of information can actually be transmitted per second using each channel, for the given input ensemble.

Solution:

To measure the actual rate of information transmission through a noisy channel, so we use the concept of mutual information, which quantifies how ~~many~~ much information the received bits convey about the transmitted bits. For a binary symmetric channel with flipping probability f , the mutual information is computed as follows.

$$I(X;Y) = H(X) - H(X|Y)$$

$H(X)$: Entropy of source.

$H(X|Y)$: conditional entropy, representing uncertainty about X after observing Y .

from Example 1:

$$P_0 = P_1 = \frac{1}{2} \quad \text{so } H(X) = 1 \text{ bit.}$$

$$f = 0.1 \quad (\text{flipping probability})$$

1). computing the conditional entropy $H(X|Y)$:

$$H(X|Y) = H(f) = -f \log_2(f) - (1-f) \log_2(1-f)$$

$$= H(0.1) = -(0.1 \log_2(0.1) + 0.9 \log_2(0.9))$$

$$\log_2(0.1) \approx -3.32193, \log_2(0.9) \approx -0.152$$

$$H(0.1) = -(0.1 * (-3.32193) + 0.9 * -0.152)$$

$$H(0.1) \approx 0.332 + 0.1368$$

$$= \underline{0.4688} \text{ bits}$$

2. Mutual Information $I(X;Y)$

$$I(X;Y) = H(X) - H(X|Y)$$

$$= 1 - 0.4688$$

$$I(X;Y) = \underline{0.5312} \text{ bits per transmitted bit.}$$

3. Actual Transmission Rate.

$$\text{Rate} = I(X;Y) * \text{transmitted bits per second}$$

$$= 0.5312 * 1000$$

$$\approx 531.2 \text{ bits per second.}$$

From Example 2:

$$P_0 = P_1 = 1/2 \quad H(X) = 1 \text{ bit}$$

$$f = 0.5$$

1. Computing the conditional entropy $H(X|Y)$

$$H(f) = -f \log_2(f) - (1-f) \log_2(1-f)$$

$$H(0.5) = -(0.5 \log_2(0.5) + 0.5 \log_2(0.5))$$

$$H(0.5) = -(0.5 * -1 + 0.5 * -1)$$

$$H(0.5) = 1 \text{ bit.}$$

2. Mutual Information $I(X;Y)$:

$$I(X;Y) = H(X) - H(X|Y) = 1 - 1 = \underline{0} \text{ bits per transmitted bit.}$$

3. Actual Transmission Rate

$$\text{Rate} = I(X;Y) * \text{transmitted bits per second.}$$

$$= 0 * 100 = 0 \text{ bits per second}$$